



Beyond technology meta-landscapes: The top 25 IPCs by landscape dynamics

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ABSTRACT

This study presents a meta-landscape analysis across the entire global patent corpus at the International Patent Classification (IPC) subclass level. We apply the concept of landscape dynamics to examine the structural evolution of the largest 549 of the 620 subclasses from 2000 to 2023. Leveraging advanced patent analytics tools, we assess key indicators such as growth rates, landscape fragmentation indices, and top assignee expectation counts to uncover patterns in technology development. Our findings uncover two trends. In a first we observe an inverse correlation between IPC fragmentation and top assignee expectation count - larger IPCs tend to be more consolidated. In a second the mean level of IPC fragmentation amongst assignees is observed to increase monotonically over the period studied. These two apparently contradictory trends are reconciled by considering the theoretical underpinnings of each. The first is an outcome of the phenomena of preferential attachment, prevalent throughout the social and biological sciences. The second is a reflection of the current state of rapid technology evolution in which fragmentation of patent ownership is driven by the entry of new actors into established but diversifying technology domains, reducing portfolio concentration among dominant patent holders. We further propose and present the results of a formalism which ranks the top 25 IPCs by a weighted, normalized combination of static and dynamic scale and fragmentation metrics. This comprehensive approach offers new insights into the shifting structure of innovation ecosystems and provides guidance for policymakers, researchers, and industry stakeholders interested in long-term technology and innovation trends.

1. Introduction

The unprecedented development and application of sophisticated data analytics tools to patent databases over the last two decades has provided business analysts, executives and public servants with new holistic insights into the evolution of technology-driven markets. In this context, holistic means the whole is greater than the sum of its parts; if one were to read every single patent abstract classified by IPC subclass G06N, i.e. computational models, (of which there are currently just over 600,000 patent families and growing at an alarming rate) this reductionist approach would not allow one to recognize, quantitatively, the dynamics of this technology landscape. In this context we use this terminology to describe an emerging domain of patent analytics defined here as the quantification of the time dependence of scale, and distribution of value, amongst patent owners in a specific technology landscape, or amongst technologies within a given assignee's portfolio.

Virtually all commercially available patent analytics tools on the

market today enable graphical visualization of the time evolution of the patent portfolio size. Such graphs are ubiquitous in patent analytics reports which indicate how technology A is outgrowing technology B or how assignee X is out-filing assignee Y. To understand the significance of changes in patent filing rates in differing technology landscapes, it is insufficient to measure only their comparative growth rates. Changes in filing rate in each landscape may be the result of differing and even competing mechanisms. In a first instance, growth may be the result of a race in which the top assignees, the incumbents, are racing away from the pack – this we may term a consolidating landscape in which more and more of the value is concentrated in the hands of fewer and fewer assignees. In a second instance, growth may be driven by the entry of a large number of challengers, resulting in a fragmented landscape in which the value is more evenly distributed amongst the assignees.

The authors assert that this information – namely the level of fragmentation of the landscape and its direction of change with time – are metrics deserving equal attention alongside the landscape size and

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growth rate but hitherto commonly ignored by patent analysts. Analogously, growth in a given assignee's patent portfolio resulting from a shotgun approach in a diversified range of technologies should be interpreted differently from growth resulting from a razor-sharp focus on a given technology. As a discipline, landscape dynamics seeks to tease out and quantify these market and assignee behaviors.

We do not restrict ourselves here to the dynamic behavior of a subset of assignees or particular technology landscapes. We have analyzed the whole corpus of patent data, which as of end 2023 comprised 15.4 million active patent families [1], providing a comprehensive overview, or what may be termed a meta-landscape, of patent information worldwide. We have, however, restricted the scope to the dynamics of technology landscapes; we will present results of a landscape dynamic analysis of the full technology meta-landscape, i.e. all patent classifications,¹ to behold a holistic view of the dynamics of the global technology patent landscape. We shall also use a range of metrics to address the question “Which are the top 25 IPCs?” Cognizant of the obvious question “by what criteria?” we shall present a range of results and then select the top 25 according to a tunable merit function using a weighting method to be described below. In the hope that the reader's curiosity is aroused, we shall leave the output of this merit function until the end of the results section; for now, we shall simply state that there are some obvious findings, but also quite a few surprises.

2. Background and theoretical framework

The use of exponents to characterize growth rates in both economics [2] and demography [3] dates back to the late 18th century. They are currently used ubiquitously in every scientific domain to quantify growth by using the slope of a log-linear plot to estimate the exponent. However, it was not until the late 19th century, that Vilfredo Pareto

showed empirically that wealth and income follow a power-law (Pareto) distribution, using log-log plots of the number of people above a given income level versus that income level to estimate the Pareto exponent [4]. The theoretical underpinnings of what became known as discrete Pareto analysis (DPA) were further developed by Udny Yule [5] in his seminal work on the distribution of species among genera. The “Yule process” is known today as preferential attachment [6], or more commonly – “the rich get richer”. This concept and associated mathematical methods were further developed and subsequently applied in numerous fields of scientific endeavour, including scientific productivity [7], linguistics [8], bibliometrics [9], information theory [10] and statistical mechanics [11].

In the realm of patent analytics, Narin's “Patent bibliometrics” paper [12] was a landmark, providing a programmatic overview that defines patent bibliometrics and how to use patents and patent citations as indicators of technological activity and science–technology linkages, but it failed to identify power laws as a tool of landscape analysis. Lotka-type power-law reasoning was later imported into inventor–patent distributions with an explicit stochastic Lotka model applied to patent data [13], with power law distributions of patents amongst assignees subsequently proposed by O'Neal & Hendy [14] as indicators of innovation at the national level. A more general (Zipfian) landscape analytical approach was subsequently developed [15] and a more rigorous best-known method for quantifying fragmentation in a patent meta-landscape was recently published [16].

Against this backdrop, our meta-landscape approach builds on this platform to provide a growth and distribution characterization model of technological competition at the level of patent ownership. Rather than focusing on a single IPC or a restricted set of assignees, we observe the joint evolution of scale (portfolio size, top-assignee expectation count) and structure (fragmentation level and trajectory) across a substantially

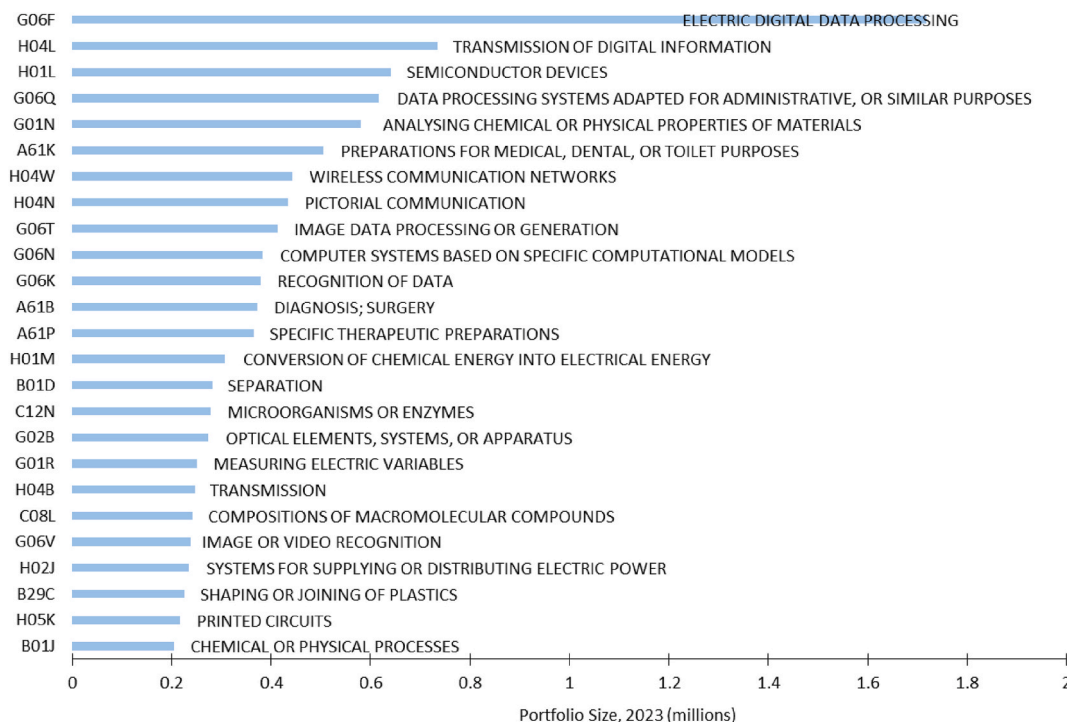


Fig. 1. Top 25 IPCs ranked by portfolio size (PS), reporting date December 31, 2023.

¹ A limited number of technology landscapes were omitted due to statistical limitations.

complete patent corpus over more than two decades. This allows us to ask whether there exist systematic “signature” trajectories of consolidation and fragmentation across technologies, and to what extent these trajectories are consistent with theoretical expectations derived from

preferential-attachment and skew-distribution models. By ranking IPC subclasses using a composite merit function that blends static and dynamic landscape metrics, we operationalize these theoretical insights into a comparative framework that identifies technology domains where cumulative advantage, entry of new players, and structural reconfiguration of patent ownership are most pronounced.

3. Method and results

3.1. IPC classification

In the analysis which follows, we restrict ourselves to technology domains as defined by WIPO's International Patent Classification (IPC), at the level of the four-digit subclass [17]. Of the 639 IPC subclasses, 549 were included in the analysis, as will be explained below. The details and trade-offs associated with this classification level choice have been previously addressed [16]. IPC was selected because of its universal global coverage and long-term stability, enabling consistent cross-country and longitudinal patent analysis. In contrast, the uneven international adoption and frequent structural revisions of the Cooperative Patent Classification (CPC) can introduce geographic and temporal biases in global studies [18].

3.2. Data curation

All data used in this publication was generated using LexisNexis PatentSight [19] an online business intelligence solution for patents. The PatentSight platform offers an extensive range of patent analytics tools but bespoke analysis, such as applying the metrics explored in this paper, are done outside of the platform using data exports from PatentSight, generously provided by the vendor for reporting dates at yearly increments from 2000 to 2023. The data export provided global coverage at the simple patent family level in more than 90 patent authorities using a number of standard PatentSight concepts, such as active patent families, the current ultimate owner, and the reporting date concept [19]. Details and trade-offs regarding why the PatentSight database was chosen for this avenue of research has been previously addressed [16].

3.3. Landscape metrics

In this section all the metrics presented in the results section will be defined. The first two metrics (portfolio size and growth rate) are well established but still require explicit definition. The subsequent metrics are generated by discrete Pareto analysis (DPA), as described below.

3.3.1. Portfolio size (PS)

Portfolio size (PS) is the count of patent families in which at least one patent family member has been classified according to the specified IPC subclass for the given reporting date. This metric was calculated for all reporting dates from 2000 to 2023 (as reported on December 31st of each year). The top 25 IPCs ranked according to this metric appear in Fig. 1.

3.3.2. Growth rate (GR)

Growth rate (GR) was calculated using the mean compound annual growth rate (CAGR) of the portfolio size over the last three years and was therefore calculated only for the most recent reporting date. The top 25 IPCs ranked according to this metric appear in Fig. 2.

3.3.3. Fragmentation index (FI)

Discrete Pareto analysis (DPA) is a mathematical method in which the distribution of resources between players in a field is characterized by the Pareto distribution, named after the Italian economist, Vilfredo Pareto [4]. The method is based on the assumption that the assignee patent family count versus rank follows a power law distribution, which may be statistically modelled by linear regression of these two parameters in log-log space. This linear model yields two parameters – the slope (i.e. the Fragmentation Index minus unity) and the vertical axis intercept (the Top Assignee Expectation Count). It was applied to 549 technology landscapes (IPC subclasses), each comprising the 100 top ranked patent assignees by portfolio size, as described in mathematical detail in Ref. [15] and according to the best-known method previously disclosed [16]. As described there, the top 100 approach was validated by consideration of three criteria, coefficient of determination, avoidance of discretization bias and weighting in favour of the top-ranked assignees. The relevant details from this reference have been reproduced in Appendix A. 71 technology landscapes in which the top ranked

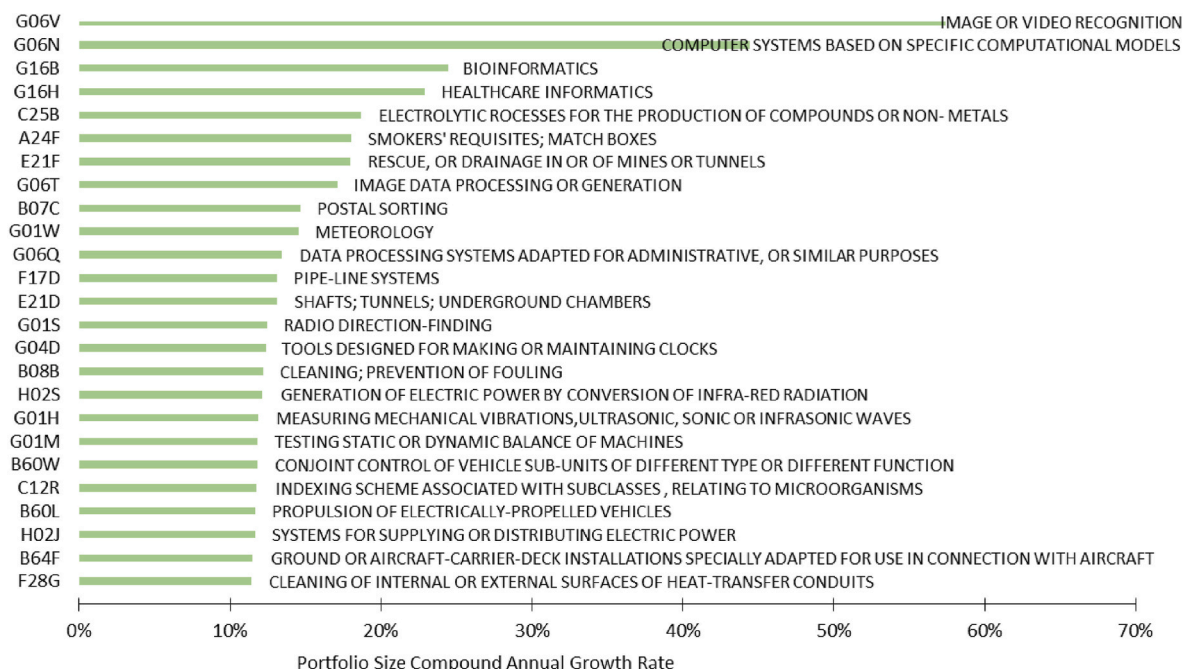


Fig. 2. Top 25 IPCs ranked by mean compound annual growth rate (GR), from 2020 to 2023.

assignee had less than 100 patent families were excluded from the analysis, due to discretization error at the bottom of the distribution where small single-digit portfolio sizes exist. Furthermore, an additional preprocessing step, not implemented in the analysis reported in previous publications, was applied to further diminish the impact of discretization which otherwise tends to bias the fragmentation index upwards for landscapes with small portfolio sizes. The mathematical details of this method can be found in [Appendix B](#). The fragmentation index (FI) presented here is a modification of the previously published consolidation index (CI) from Ref. [16]; it is simply defined as $1 - CI$. This metric was calculated for all reporting dates from 2000 to 2023. The top 25 IPCs ranked according to this metric appear in [Fig. 3](#).

3.3.4. Top assignee expectation count (C_1)

While this is also a metric of scale, in contrast to portfolio size (PS) which is an indicator of the size of the technology landscape as a whole, the top assignee expectation count (C_1) is an indicator of the size of the portfolio of the top ranked assignee. We consider this metric more informative than the portfolio size, because it provides an indication of the scale of the major players in the technology landscape, which cannot be ascertained from portfolio size only, as it depends on the level of consolidation. The mathematical details of its calculation can be found in Ref. [16]. It is worth noting that this is not the actual portfolio size of the top ranked assignee; it is a regression parameter extracted from the DPA which reflects all the top assignees, and not just the top-ranked assignee, making it statistically robust. The top 25 IPCs ranked according to this metric appear in [Fig. 4](#).

3.3.5. Fragmentation trajectory (FT)

While the fragmentation index (FI) is a static metric of portfolio fragmentation for a given reporting date, the fragmentation trajectory (FT) indicates the rate of change in the fragmentation index over the last three years. It was calculated only for the most recent reporting date. The top 25 IPCs ranked according to this metric appear in [Fig. 5](#).

3.4. Meta-landscapes

A visualization which enables comparison of multiple technology landscapes has previously been termed a meta-landscape. While this term can be applied to a plethora of landscape metric combinations, in this publication we have chosen one example to illustrate the principle.

In this example we plot fragmentation index (FI) vs the top assignee expectation count (C_1) on a log scale, which are the two regression parameters extracted by DPA.

3.5. Landscape Merit Function

Now that we have defined a series of landscape metrics, we propose a method to generate a single Landscape Merit Function (LMF) which is a weighted combination of the above. In a first step each metric is independently normalized using the standard score normalization method of subtracting the mean value and dividing by the standard deviation of the metric distribution across all IPCs. In a second step all normalized metrics for a given IPC are multiplied by a user-selected weight and combined arithmetically. The weighting function in the current analysis was 0.3 for the static metrics and 0.2 for the dynamic metrics, that is:

$$LMF = 0.3 \cdot (\log_{10} PS + FI) + 0.2 \cdot (GR + FT)$$

This selection was based on statistical rather than strategic considerations; the dynamic metrics, Growth Rate (GR) and Fragmentation Trajectory (FT) displayed more skew distributions than the static metrics, Portfolio Size (PS) and Fragmentation Index (FI). In the final step, the IPCs are ranked to generate the rankings displayed in [Fig. 8](#). A cursory sensitivity analysis, in which the weighting function was equalized between all metrics resulted in a modified order but had a limited impact on the inclusion of the IPCs themselves, replacing H04L, transmission of digital information with C12H, pasteurization and sterilization.

4. Discussion

4.1. Meta-landscape dynamics

The current implementation of the discrete Pareto analysis which generates FI and C_1 for each IPC considers only the top 100 assignees by patent family count and not the long distribution tail which includes a large number of single patent owners. In this context, “new entrant” should be interpreted as an entrant who has made a substantive investment in patent filing to enter the race where the scale of the investment varies from approximately 5 to 3000 patent families for the 100th ranked assignee, depending on the scale and fragmentation of the IPC. As previously mentioned, the lower bound was used to exclude a

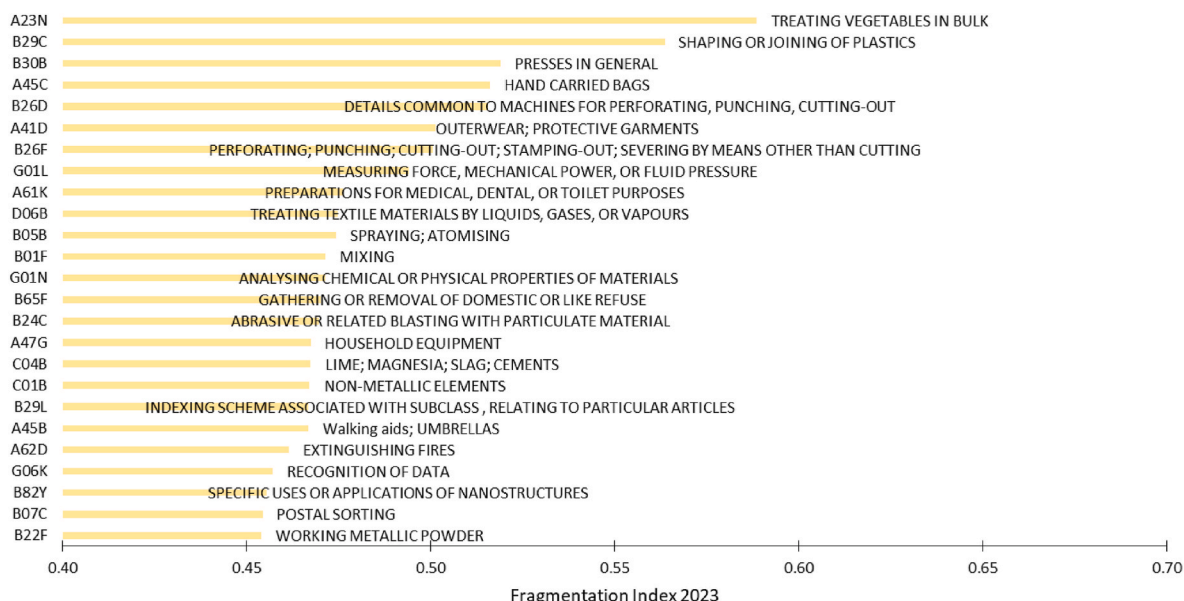


Fig. 3. Top 25 IPCs ranked by Fragmentation Index (FI), reporting date December 31, 2023.

limited number of IPCs where discretization bias renders the Pareto analysis unreliable. While this exclusion comprises 14% of all IPC subclasses it amounts to only 0.75% of all patent families, suggesting that the exclusion's impact on the representativeness of the analysis is marginal.

This meta-landscape analysis approach relies on the previously established notion that industries (and IPC subclasses) display unique 'signature' trajectories which can be useful for predicting market trends, since patenting activity can act as an early indicator that distinguishes between market growth due to leader dominance and that due to new player entry. Fig. 7 shows that the mean fragmentation index (FI) increases from 0.05 to approximately 0.2 between 2000 and 2023, while the mean Top Assignee Expectation Count (C_1) grew somewhat from 2000 until 2008 but then the trend reversed and shrunk until 2011 when the upward trend restarted, perhaps as the global economy recovered from the Great Recession. We interpret the mean Fragmentation Index upward trend with time as a possible indicator of increasing innovation since the entry of new players can encourage competition leading to inventive diversity. In such environments, new entrants regularly try to differentiate themselves, often leading to novel ideas and unique approaches to problem-solving. A lack of market dominance by a few large firms can ensure that innovation is not stifled by entrenched interests seeking to protect their existing technologies or market position. Furthermore, it has long been asserted [20,21] that fragmented industries provide greater opportunity for collaboration and the cross-pollination of ideas, as diverse players experiment and share insights, creating a dynamic ecosystem that drives technological progress and economic growth.

A second, non-contradictory observation is that the mean Fragmentation Index as a function of C_1 slopes downwards in Fig. 6. This is evident irrespective of time. So, while the mean level of fragmentation of the whole meta-landscape is increasing over time, when technology domains grow significantly (note the log scale), fragmentation diminishes, as larger players appear to consolidate their hold on more established markets. This dynamic may be understood in the framework of the well-established principle of preferential attachment [5], a mechanism where entities accrue advantages proportional to their existing resources, which has become a cornerstone in understanding power-law distributions across biological, social, and technological

systems [9] and forms the scientific basis of the analysis in this paper.

At the more abstract philosophical level, the downward trend in Fig. 6 may also be interpreted as support for Marx's famous assertion "Monopolists are made from competition; competitors become monopolists" [22]. Of course, what Marx failed to recognize is what Christensen posited 150 years later and what every contemporary MBA student learns - that economic and technological developments do not follow monotonic trajectories [23]. Market leaders are often displaced by new entrants who are able to disrupt markets by rapid product iteration or targeting overlooked emerging segments.

4.2. Interpretation and decision relevance of the Landscape Merit Function

While the Landscape Merit Function (LMF) is mathematically defined as a weighted combination of normalized scale and dynamic landscape metrics, its primary value lies in its interpretability for managerial decision-makers and policymakers. Conceptually, LMF should be viewed as a composite technology intelligence indicator that captures not only the absolute size of a technological domain, but also its growth dynamics and structural evolution. Users of the merit function may choose to modify the weighting function in accordance with their ranking perspective.

From a managerial perspective, given the current weighting, high LMF values can be considered to highlight technology domains combining large or rapidly growing patenting activity with meaningful shifts in competitive structure. Such domains may signal areas of strategic importance for corporate R&D investment, portfolio diversification, merger and acquisition activity, or strategic partnerships. In contrast, low LMF scores may indicate either technologically mature and consolidated domains, where incremental innovation dominates, or early-stage domains lacking sufficient scale to justify major strategic commitments.

From a technology policy perspective, LMF enables rapid screening of large technology spaces to identify emerging or reconfiguring domains that warrant closer qualitative assessment. In particular, high growth rates combined with strongly positive fragmentation trajectories often appear to indicate the entry of new players or the convergence of previously distinct technological trajectories. Convergence in this

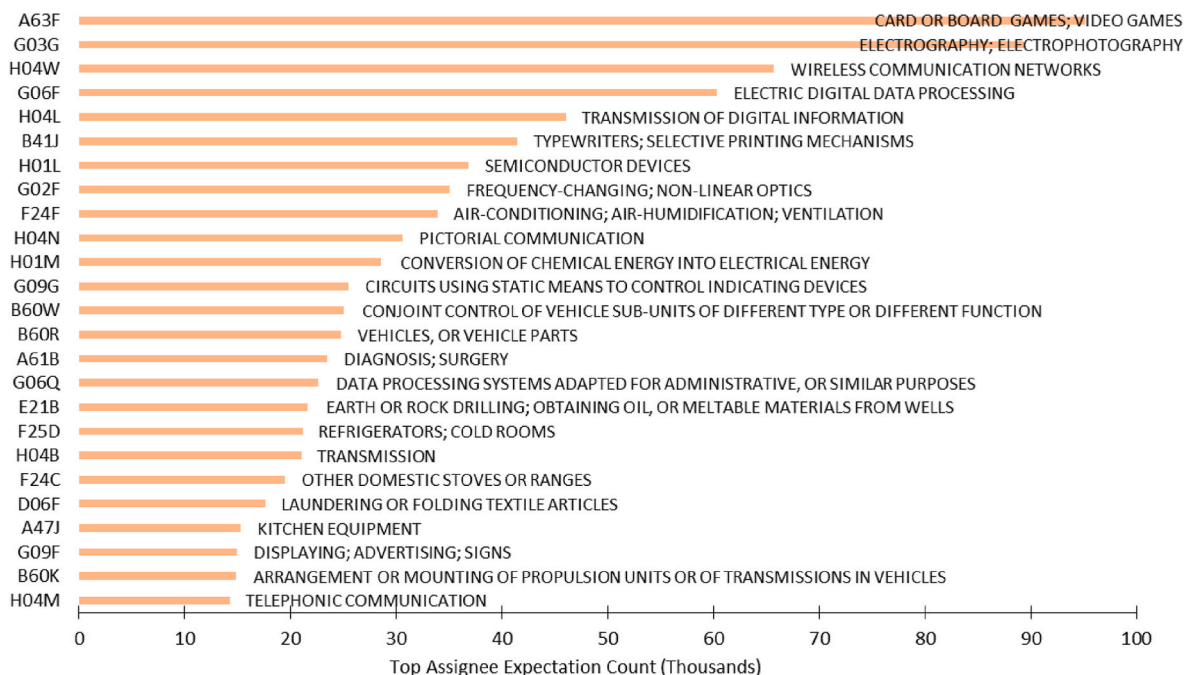


Fig. 4. Top 25 IPCs ranked by Top Assignee Expectation Count (C_1), reporting date December 31, 2023.

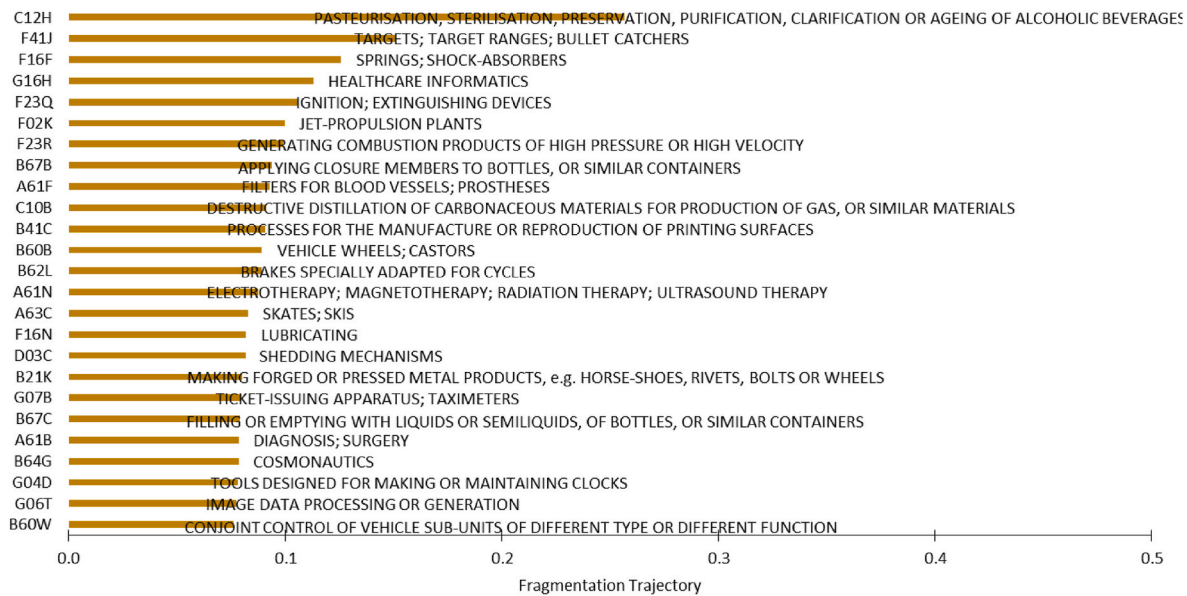


Fig. 5. Top 25 IPCs ranked by fragmentation trajectory (FT), 2020–2023.

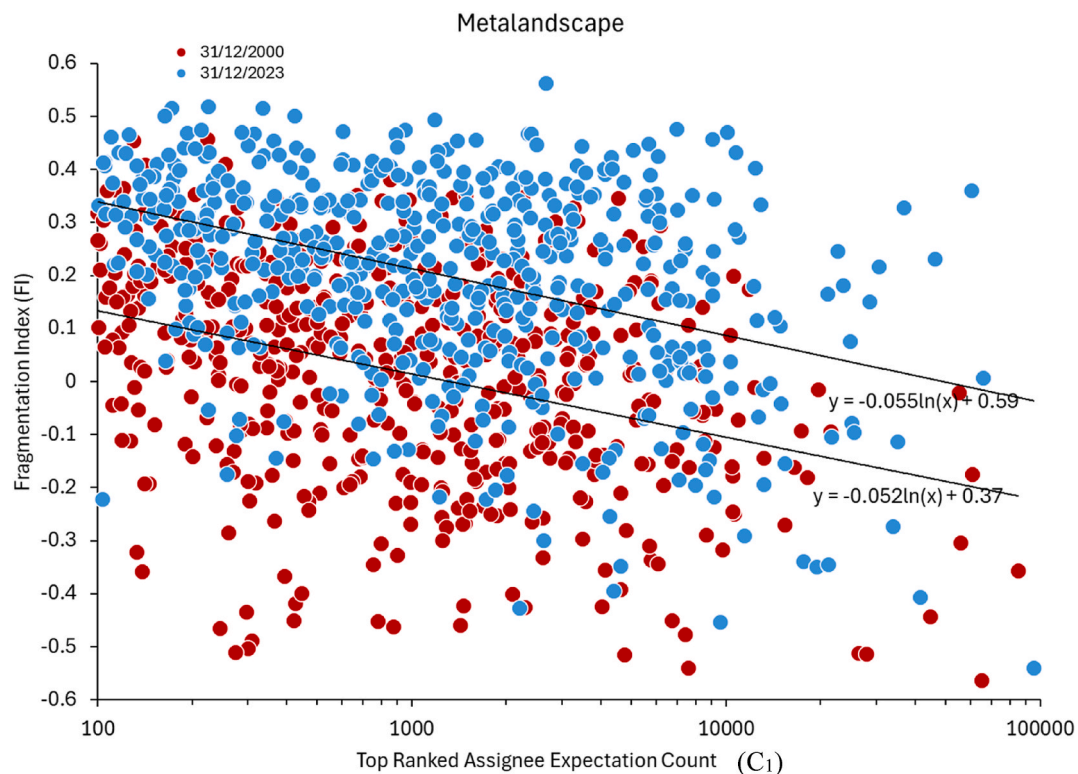


Fig. 6. Meta-landscape of largest 549 out of the 630 IPC subclasses, reporting dates December 31, 2000 (red) and December 31, 2023 (blue).

context refers to situations where technologies originating in separate domains – such as digital computing and healthcare, advanced materials and energy systems, or sensing technologies and mechanical engineering – begin to overlap and recombine within the same IPC subclass. Such recombination typically manifests as the entry of firms whose historical patenting activity lay outside the focal domain, increasing both growth and assignee diversity (increasing fragmentation). Both new entry and technological convergence are commonly associated with technological novelty and early-stage innovation. Policymakers may therefore consider using LMF and its constituent metrics to support horizon

scanning, priority-setting exercises, public research funding allocation, and the identification of candidate domains for targeted innovation support instruments.

More generally, LMF provides a systematic, data-driven mechanism for distinguishing between technology growth by incumbent consolidation and growth driven by the entry of new innovators. This distinction is critical for both corporate and public-sector stakeholders, as these two growth modes imply fundamentally different innovation dynamics, competitive environments, and policy implications.

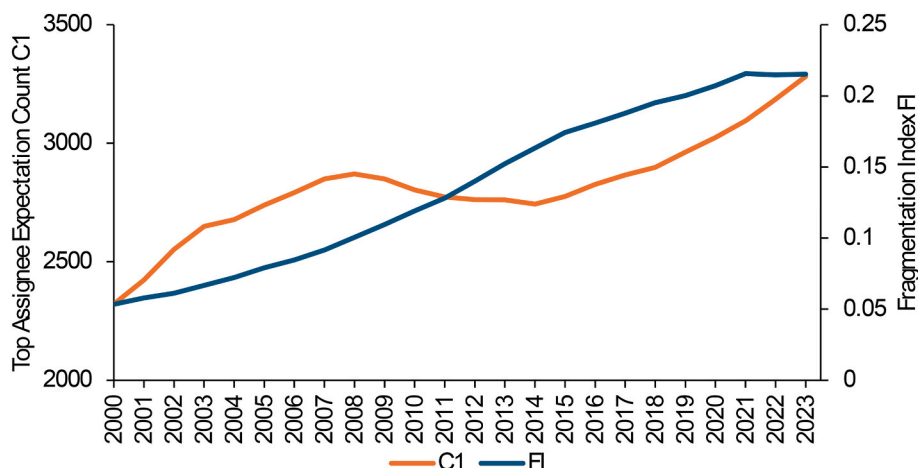


Fig. 7. Time evolution of mean landscape metrics, Top Assignee Expectation Count (C_1) and Fragmentation Index (FI), 2000–2023.

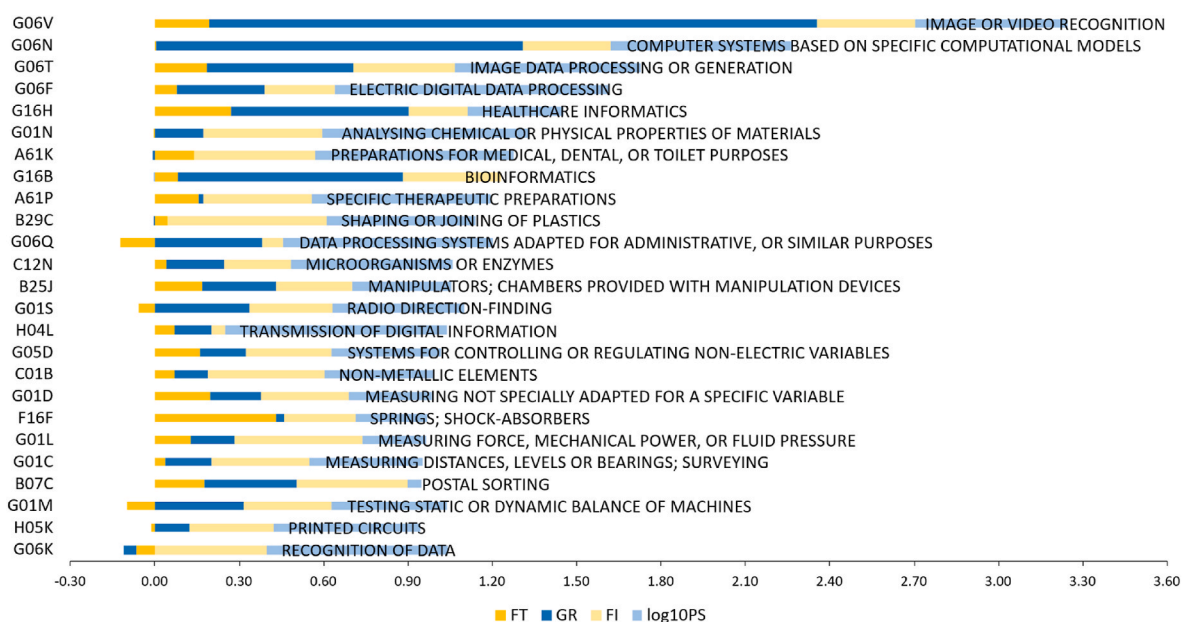


Fig. 8. Top 25 IPCs by Landscape Merit Function, reporting date December 31, 2023.

4.3. Industry specific insights from top 25 merit ranking

To draw insight from the IPC subclass ranking results, we shall first break down the top 25 into similar technology domains about which broad generalizations may be made. We shall then identify known industrial growth trends associated with these technology domains.

4.3.1. G06 class – computing, calculating or counting

There are no less than 6 subclasses in the top 25 rankings in this class, including the top 3. It is worth noting that the top ranked subclass G06V (image or video recognition) outperforms even G06N (computational models) which encompasses a substantial proportion of Generative AI inventions. This is somewhat inconsistent with recent industry reports [24] which place Generative AI and Natural Language Processing (for which the dominant portion falls in G06F (computing)), as both larger

and faster growing AI segments than image or video recognition. Looking more closely at the top two, what separates them is G06V's GR (the highest of all subclasses) and its positive fragmentation trajectory, indicating that much of the growth is due to new entrants.² A possible explanation for the disparity between the above-mentioned industry market analysis pointing towards G06N and the patent analytical ranking, placing G06V at the top may be the delay between patent portfolio dynamics and market dynamics. Alternately this may be the result of differing filing and prosecution policies for image recognition vs generative AI and LLM in the board room and the patent office, respectively. In 3rd place is G06T (image data processing or generation) which displays high scores roughly equally distributed between all four metrics. In this subclass, we find use cases related to entertainment, media, autonomous vehicles, facial recognition, diagnostic imaging and telemedicine, cutting across industry boundaries.

² Robustness tests confirmed that mass automated backclassification of newly created IPC subclasses – such as G06V, introduced in 2021 - did not impact these results. Backclassification is designed to make statistical analysis comparable across time, even for periods before the subclass formally existed.

4.3.2. G16B and G16H subclasses – bio and healthcare informatics

The G16B and G16H subclasses both appear in the top 10 ranking. Both subclasses fall under the broader trend of digital transformation (AI, big data, IoT) in healthcare and life sciences. G16H relates to ICT systems used in healthcare delivery, and includes electronic health records, telemedicine, clinical decision support, AI diagnostics, and remote patient monitoring. G16B, on the other hand, focuses on using computing to analyze biological data, such as genomics and proteomics, and is likely to be driven by recent growth in personalized medicine and synthetic biology.

4.3.3. G01 class – measuring or testing

There are 5 subclasses in the G01 class in the top 25. Ranked 6th overall is G01N, analyzing chemical or physical properties. This is a broad category for patents in scientific instrumentation, laboratory analysis, and industrial quality control covering bio-medical, environmental and industrial use cases. Next is G01S ranked 14th, which covers positioning, navigation, and guidance systems, satellite and terrestrial which cut across automotive, aviation and mobile. It is interesting to note that this IPC displays a negative fragmentation trajectory indicating growth through industry consolidation. At the core of the G01 class is metrology, the act of creating value by measuring it. This domain is set to continue to flourish as a foundational element of 21st century technology and industry.

4.3.4. B29C and B22F ((shaping plastics and working metallic powders)

The inclusion of B29C (shaping plastics) and B22F (working metallic powders) amongst the top 25 IPC subclasses highlights the accelerating impact of 3D printing (additive manufacturing) technologies on global innovation landscapes. B29C covers injection molding, extrusion, thermoforming, and plastic welding – all used in industries including aerospace, automotive, medical devices, and consumer goods – whereas metallic additive manufacturing enables the production of complex, lightweight, and high-strength components, especially for aerospace, medical implants, and industrial tooling.

4.3.5. A61 class – medical and veterinary science

The A61 class lies at the heart of pharmaceutical, medical device, and healthcare innovation, so it is not too surprising to see three subclasses within this class in the top 25 ranking. A61K (Preparations for medical, dental, or hygienic purposes), A61P (Therapeutic activity of chemical compounds or medicinal preparations) and A61B (Diagnosis; Surgery; Identification) all appear in the top 20. A61K is one of the largest IPC subclasses in the entire patent system, and nearly every pharmaceutical patent lands here. A61P complements A61K by classifying therapeutic effects (e.g. anti-cancer, anti-inflammatory) whereas A61B covers medical devices, imaging, diagnostics, and increasingly AI-driven tools. Large IPC subclasses such as these do not usually exhibit strong growth, due to large absolute size, but the increasing fragmentation in these areas suggests strong growth in the sector with new players flooding in due to expanding global healthcare needs (aging populations, chronic diseases), new modalities (such as mRNA therapies), digitization and AI in diagnostics, and COVID-19 pandemic-driven R&D.

4.3.6. F16F subclass – shock absorbers

This classification has made it into the ranking by virtue of a high fragmentation trajectory – a rapid influx of new players. The reason for this influx is that F16F contains many new shock-absorber architectures, in particular what are termed active suspension systems which use electronic sensors and computer-controlled actuators to dynamically adjust damping in real-time, providing a smoother ride and improved handling compared to passive suspension systems. The increased and redistributed mass of battery powered EVs is the key driver behind these emerging requirements.

4.3.7. Other subclasses

Exploring the other IPC subclasses within the top 25, we see the inclusion of C12N (microorganisms or enzymes; genetic engineering) which lies at the heart of biotechnology and covers genetically-modified organisms, engineered cells, CRISPR tools and the production of proteins, enzymes, vaccines (e.g. COVID-19 mRNA vaccines) and antibodies. Although A41D (outerwear; protective garments) is not traditionally viewed as high-tech, its inclusion is largely due to personal protective equipment (PPE) such as masks, gowns, hazmat suits (an artefact of R&D activity relating to the COVID-19 pandemic), as well as smart textiles (garments with sensors and electronics) and military and industrial wearables with integrated functions.

4.3.8. Limitations and generalizability

The present analysis is subject to several limitations that should be considered when interpreting the results and their possible extensions. First, all metrics are derived from a single commercial patent database and a specific set of harmonization rules (e.g. ultimate owner definitions, active family criteria, treatment of continuations and divisionals). Although PatentSight offers broad international coverage and well-documented ownership cleaning [25], alternative data sources or disambiguation strategies could yield somewhat different portfolio sizes and assignee rankings, particularly in jurisdictions with complex corporate structures or limited legal-status information. Second, our implementation of discrete Pareto analysis deliberately focuses on the top 100 assignees per IPC, excludes subclasses with very small C_1 values and applies a particular interpolation scheme to mitigate discretization bias in the lower tail. These choices are empirically motivated and validated in previous work, but they inevitably down-weight the contribution of long-tail, small-portfolio players and may understate fragmentation in very nascent or niche technology domains. Conversely, the lack of reliable name harmonization for (apparently) single patent owners may result in an artificially extended tail of the patent family distribution, *overestimating* fragmentation.

A further limitation arises from the use of IPC subclass classifications and patent families as stable proxies for technological domains and underlying inventive activity. IPC assignments can lag actual technological developments, may differ systematically across patent offices, and do not capture all forms of innovation (e.g. trade secrets, industrial designs or open-source contributions). Similarly, using active patent families at fixed year-end reporting dates captures a particular legal and temporal slice of technology dynamics that may not align perfectly with market adoption or R&D investment cycles, especially in rapidly evolving domains such as digital technologies. These factors constrain the direct generalizability of our results to non-patent measures of innovation and to alternative classification frameworks such as CPC or bespoke technology concordances. Nonetheless, because the core findings concern relative patterns—such as the inverse relationship between fragmentation and top-assignee scale, and the upward trend in mean fragmentation across a very broad set of IPC subclasses—we expect the qualitative insights on consolidation versus fragmentation, and on the value of meta-landscape perspectives for comparative technology intelligence, to be robust to reasonable variations in the choice of database and taxonomy.

5. Conclusions

This study demonstrates the value of a meta-landscape analysis approach to patent data, revealing consistent and interpretable patterns in technological evolution across time. By examining IPC subclass dynamics, we can distinguish between growth driven by incumbent dominance versus the entry of new innovators. From 2000 to 2023, a notable rise in the mean Fragmentation Index (FI) signals increasing diversification and a decline in market consolidation, pointing to an influx of new players and suggesting a more competitive, innovation-rich environment. Moreover, while the mean Top Assignee

Expectation Count (C_1) reflects fluctuations aligned with broader economic cycles, the inverse relationship between FI and C_1 remains stable over time. This suggests that as specific technology domains mature and scale, they tend to become less fragmented, with leading firms consolidating their positions in a drive towards dominance according to the established mechanism of preferential attachment.

The meta-landscape approach provides a framework for analyzing the entire global patent corpus over time, offering a holistic view of how patenting activity has evolved since the turn of the millennium. Unlike traditional analyses that focus on specific technologies or regions, this approach enables a comprehensive, system-level understanding of innovation dynamics across all technology fields and industrial sectors, offering insights for policymakers, researchers, and industry leaders seeking to interpret long-term trends in global technological development and innovation behavior.

Finally, by combining established and novel metrics to characterize technology sectors, we have presented the results of a methodology using a proposed Landscape Merit Function (LMF) which ranks IPCs according to their meta-landscape trajectories, identifying the most dynamic domains which may be set to drive global economic growth in the near to medium term future. By translating complex patent landscape dynamics into a single interpretable indicator, the LMF provides both policymakers and industry decision-makers with a practical tool to consider using for early detection of emerging technologies, strategic prioritization, and evidence-based innovation policy design.

APPENDIX A. DISCRETE PARETO ANALYSIS

The application of DPA to patent analytics has been described in detail in Ref. [15,16]. In this appendix the basic mathematics and best-known method for its application to patent analytics will be summarized.

The distribution of a random variable $C(R)$, in this case the count of patent families owned by a given assignee of rank R , where the players are ranked in descending order of patent family count, may be described by the expression:

$$C(R) = C_1 R^{(1-FI)} \quad (1A)$$

in which C_1 is a scale parameter representing the expectation value of the patent family count of the top ranked assignee and FI is a shape parameter, interpreted as a fragmentation index of the patent landscape. By applying a base 10 logarithm to equation (1A) above, the following linear expression

$$\log_{10} C(R) = \log_{10} C_1 - S \log_{10} R \quad (2A)$$

is obtained where the slope, $S = FI-1$. For a given patent landscape with observed patent count distribution $C_1 \dots C_n$ for ranked assignees 1 to n , the system is overdetermined, and we apply a simple least squares method to determine the estimators for the exponent and top ranked assignee patent count. A vanishing Fragmentation Index, that is $FI = 0$, is the special case known as Zipf's Law [8] where the exponent is minus unity. This conveniently subdivides landscapes into consolidated when $FI < 0$ and fragmented when $FI \geq 0$. In the current analysis, n is set to 100, that is only the top 100 ranked assignee patent family counts are included in the model because these assignees are typically fully name harmonized in PatentSight and the top ranked assignees are of most commercial interest. The accuracy limitations and goodness of fit data of the method have been investigated and are described in detail in Ref. [16].

APPENDIX B. DISCRETIZATION BIAS AND ITS TREATMENT

For patent landscapes with small portfolio sizes the DPA method suffers from discretization bias which biases FI upwards. The root cause of this inaccuracy is the low patent count tail where multiple assignees have identical single digit patent counts which diminishes the slope of the regression. This effect is exacerbated in the case of highly consolidated landscapes. It may be mitigated using a simple linear interpolation of the rank parameter as follows. If there are m assignees with equal patent family count, then the rank R is replaced with

$$R' = R - \frac{i}{m} \text{ for } i = 1 \text{ to } m \quad (1B)$$

The effect of this correction is shown in the attached synthetic data set in Table 1.

Table 1

Synthetic DPA data on a log-log scale, before application of the preprocessing step (red) and after (green). All cases have the top assignee expectation count C_1 set to 100. The Fragmentation Index FI is set (left to right) to 0.4, 0 and -0.4 respectively. The regression results are also shown in the top right.

CRediT authorship contribution statement

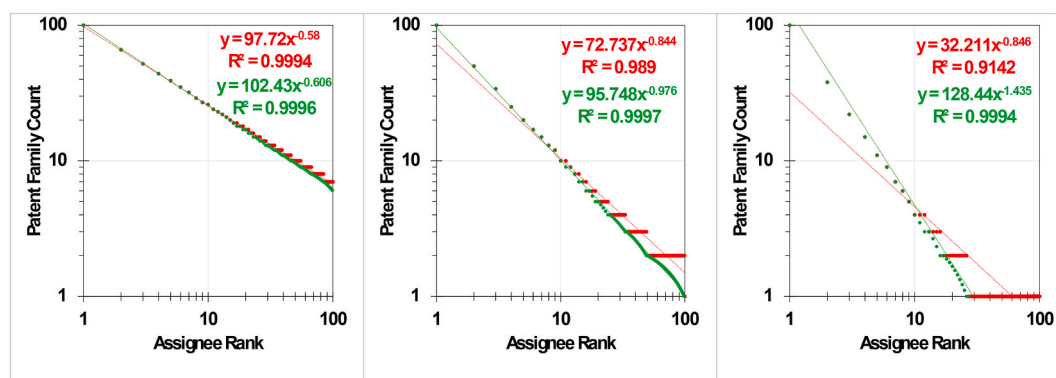
Michael E. Adel: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christopher Harrison:** Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Upon review of this data, it is evident that the match to the model is improved in all cases by the preprocessing (recall that the exponent, $S = FI-1$). For fragmented landscapes such as the case on the left, the effect is marginal, but for the consolidated landscape on the right the improvement in the match with the synthetic result ($C_1 = 100$ and $FI = -0.4$) is substantive. Furthermore, the coefficient of determination (R^2) is improved in all cases. However, for cases with C_1 less than 100, the match to the synthetic data continues to degrade and so patent landscapes with $C_1 < 100$ have been excluded from the analysis, as already mentioned. It is noted that the rank $\frac{1}{2}$ method [26] was also tested, but for the numerical characteristics of the current data, the above-described interpolation method provided a better approximation in the ranges of relevance for C_1 and FI .

Data availability

Data will be made available on request.

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